Pintle Injector Project Checklist — Spiral + Vane Manifold

A phase-by-phase roadmap from first decision to final report, plus a master list of every parameter you'll need to pin down along the way. Check items off as you go; treat Phase 0 as a single sitting, everything after as iterative.

Phase 0 — Foundational decisions (decide before calculations or CAD)

- [ ] Define project goal: cold-flow demonstrator only, or working toward hot-fire?
- [ ] Choose propellant pair (or confirm cold-flow simulant — typically water for one side, air or water for the other)
- [ ] Decide oxidizer-centered vs. fuel-centered pintle
- [ ] Set target thrust (or equivalent target volumetric flow rate, if cold-flow only)
- [ ] Set target chamber pressure (hot-fire) or supply/test pressure (cold-flow)
- [ ] Decide required throttling ratio (e.g., 3:1, 5:1, 10:1)
- [ ] Decide manufacturing method/access (CNC, metal 3D print, resin 3D print, student machine shop)
- [ ] Decide test method: visualization only, CFD-validated, or instrumented cold-flow rig
- [ ] Write down your exact research question and what "success" looks like (this drives Phase 12's analysis)
- [ ] Confirm budget/timeline covers building **two** manifold versions — baseline and spiral+vane — since the whole project rests on that comparison

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Phase 1 — Performance & flow calculations

- [ ] Run NASA CEA (or equivalent) for combustion properties if hot-fire is the goal — skip if cold-flow only
- [ ] Calculate total propellant mass flow rate for your target thrust/pressure
- [ ] Split mass flow into oxidizer and fuel streams using your O/F ratio
- [ ] Calculate required flow areas (annular gap area, radial tip gap area) from mass flow, density, and target injection velocity
- [ ] Assume/estimate a discharge coefficient (Cd) for your geometry
- [ ] Set a pressure-drop budget across the injector (rule of thumb: ~15–25% of chamber pressure for hot-fire; pick an equivalent for cold-flow)
- [ ] Estimate spray velocity, Weber number, and expected spray angle as an atomization sanity check

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Phase 2 — Core pintle/sleeve geometry

- [ ] Pintle tip diameter (Dt)
- [ ] Pintle rod diameter (Dp)
- [ ] Pintle angle (α)
- [ ] Annular gap thickness (tag) at nominal throttle position
- [ ] Pintle gap thickness (tpg)
- [ ] Tip thickness (tt)
- [ ] Shadow angle (θ), if using a slotted tip
- [ ] Slot count/pattern at the tip (if slotted rather than a continuous gap)
- [ ] Total momentum ratio (TMR) check between oxidizer and fuel streams for correct impingement

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Phase 3 — Throttling range & mechanism

- [ ] Opening distance range: Laod_min (closed) to Laod_max (full open)
- [ ] Actuator type (stepper + lead screw / manual micrometer / pneumatic)
- [ ] Actuator travel, resolution, and force requirements
- [ ] Actuator-to-rod coupling design
- [ ] Guide bushing count and axial locations (minimum two — single-point support causes bending/wobble)
- [ ] Bushing material and clearance fit
- [ ] Dynamic seal type and groove dimensions where the rod passes through the body
- [ ] Seal material compatibility with your propellant/simulant
- [ ] Confirm the actuator's position doesn't block the central oxidizer feed path — route around it if needed

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Phase 4 — Manifold redesign: spiral cavity

- [ ] Decide which propellant path needs the spiral fix — outer fuel manifold, and possibly also the central oxidizer feed if the actuator forces it to be side-fed too
- [ ] Choose your target uniformity metric: constant velocity, or constant static pressure, around the circumference
- [ ] Derive the area-vs-angle profile A(θ) from mass conservation (area proportional to flow remaining at each angular position)
- [ ] Set starting radius (at inlet, θ = 0°) and ending radius (at θ = 360°, before the vane ring)
- [ ] Design the "tongue" transition where the spiral's start and end nearly meet
- [ ] Decide the tangential inlet duct geometry and entry angle
- [ ] Check channel width against boundary-layer thickness at your Reynolds number (this is your scale-effect sanity check — turbine volutes are meters across, yours is millimeters)

- [ ] Decide manufacturing approach for the spiral cavity (very likely 3D printing given the geometry)

Phase 5 — Manifold redesign: turning vanes

- [ ] Vane count around the circumference
- [ ] Vane radial position (inner/outer radius of the vane zone)
- [ ] Vane angle/orientation
- [ ] Vane thickness and length (balance swirl removal against added pressure drop/blockage)
- [ ] Check vane-induced pressure drop against your Phase 1 budget
- [ ] Decide vane manufacturing approach (integral to the 3D print, or a separate inserted part)

Phase 6 — Baseline design (your control case)

- [ ] Model a baseline single-side-inlet manifold with no spiral and no vanes
- [ ] Keep every other dimension (Dt, tag, Dp, α, etc.) identical to the spiral+vane version — the manifold must be the *only* variable that changes between the two, or your comparison won't be valid

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Phase 7 — SolidWorks modeling (parametric)

- [ ] Set up a global variables/equation sheet covering every Phase 2–5 parameter
- [ ] Model the pintle rod (revolve + shell + tip slots if used)
- [ ] Model the sleeve/body with the inner diameter linked by equation (Dt + 2×tag)
- [ ] Model the spiral manifold cavity per your Phase 4 profile
- [ ] Model the turning vanes per Phase 5
- [ ] Model the baseline manifold as a separate part/configuration per Phase 6
- [ ] Model actuator, coupling, bushings, and seals
- [ ] Assemble with concentric mates + a travel-limited distance mate for throttle motion
- [ ] Create configurations or a design table covering throttle position (closed/mid/open) × manifold type (baseline/spiral+vane)

Phase 8 — Verification

- [ ] Interference detection across the full throttle travel range
- [ ] Flow Simulation (or external CFD) comparing baseline vs. spiral+vane manifold

- [ ] Quantify circumferential uniformity numerically in the CFD results (e.g., velocity or pressure variation sampled around the exit ring)
- [ ] Structural sanity check (wall thickness vs. expected pressures) if hot-fire is a goal
- [ ] Add dimension sensors to flag if tag ever drops below a safe minimum as you tune other parameters

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Phase 9 — Manufacturing

- [ ] Finalize manufacturing method per part (rod, sleeve+manifold, vanes, bushings)
- [ ] Check minimum feature size against your manufacturing method's real capability — especially the spiral cavity and vane thickness
- [ ] Order/print/machine all parts
- [ ] Inspect parts against CAD dimensions, especially tag, since it drives everything downstream
- [ ] Assemble and dry-fit; confirm the actuator moves the rod smoothly through its full range with no binding

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Phase 10 — Test setup

- [ ] Build the cold-flow rig (pump or pressurized tank, flow meters, pressure gauges)
- [ ] Set up supply lines for your simulant, with equal-length symmetric branching if your design needs balanced flow
- [ ] Set up visualization — camera/shadowgraphy for spray pattern, or dye injection for internal flow
- [ ] Set up pressure taps around the manifold circumference if you want to quantify uniformity directly, not just visually
- [ ] Calibrate flow meters and actuator position readout

Phase 11 — Testing

- [ ] Run the baseline manifold at multiple throttle positions; record spray pattern/uniformity
- [ ] Run the spiral+vane manifold at the same throttle positions, same conditions
- [ ] Repeat runs to check repeatability
- [ ] Record flow rate vs. throttle position (your throttle curve) for both configurations

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Phase 12 — Analysis

- [ ] Quantify spray/flow uniformity for both configurations (e.g., angular variation in spray

angle or exit flow rate)
- [ ] Compare experimental results against your Phase 8 CFD predictions
- [ ] Compare throttle-curve repeatability between the two configurations
- [ ] Identify and document any failure modes or unexpected behavior (leaks, cavitation, actuator binding, etc.)

Phase 13 — Documentation

- [ ] Write up methodology, citing your design equations and sources
- [ ] Present CAD drawings, including the family-of-positions (closed/mid/open) drawing
- [ ] Present results — uniformity comparison, throttle curves, CFD-vs-experiment agreement
- [ ] Discuss limitations and future work (scale effects, hot-fire extension, more vane/inlet configurations)

Master Parameter List

Every factor you'll need to pin down, grouped by category, for quick reference while you work.

Mission & requirements

Parameter	Notes
Target thrust	Or equivalent flow target for cold-flow
Chamber / supply pressure	
Throttle ratio	e.g. 3:1, 5:1, 10:1
Propellant pair / simulant	
O/F ratio	

Propellant flow

Parameter	Notes
Total mass flow rate	
Oxidizer mass flow rate	
Fuel mass flow rate	

Parameter	Notes
Propellant density (each)	
Propellant viscosity (each)	Matters for Phase 4 scale-effect check
Injection velocity (each stream)	
Discharge coefficient (Cd)	
Injector pressure drop budget	

Pintle/sleeve geometry

Parameter	Notes
Pintle tip diameter (Dt)	
Pintle rod diameter (Dp)	
Pintle angle (α)	
Annular gap thickness (tag)	Also your throttle variable
Pintle gap thickness (tpg)	
Tip thickness (tt)	
Shadow angle (θ)	If slotted tip
Slot count/pattern	If slotted tip

Throttling mechanism

Parameter	Notes
Opening distance range (Laod min/max)	
Actuator type	Stepper/lead screw, micrometer, pneumatic
Actuator travel, resolution, force	
Guide bushing count & location	Minimum two
Bushing material/fit	
Seal type & groove dimensions	
Seal material compatibility	

Spiral manifold

Parameter	Notes
Target uniformity metric	Constant velocity vs. constant pressure
Starting radius (θ=0°)	
Ending radius (θ=360°)	
Area-vs-angle profile A(θ)	Derived from mass conservation
Tangential inlet duct geometry/angle	
Tongue transition design	
Channel width vs. boundary layer thickness	Scale-effect check

Turning vanes

Parameter	Notes
Vane count	
Vane radial position	
Vane angle/orientation	
Vane thickness & length	
Vane-induced pressure drop	Check against budget

Structural / material

Parameter	Notes
Wall thickness (body, sleeve)	
Material choice	Compatibility + manufacturing method
Pressure rating / safety factor	If hot-fire
Thermal considerations	If hot-fire

Manufacturing

Parameter	Notes
Manufacturing method per part	CNC / 3D print / shop
Minimum feature size achievable	Especially spiral cavity, vanes
Tolerances	Especially tag

Test & instrumentation

Parameter	Notes
Flow meter range/accuracy	
Pressure tap locations	If quantifying manifold uniformity directly
Camera/shadowgraphy setup	
Actuator position readout	
Number of repeat runs	For repeatability